MIL-S-11153F
20 August 1985
SUPERSEDING
MIL-S-11153E
12 December 1977

MILITARY SPECIFICATION

SLED, SCOW-TYPE, PLASTIC ,

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 200-pound (90 kilogram) capacity, scow-type, molded plastic sled, equipped with a cotton duck cover.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping Steel, and Seals.

MILITARY

MIL-T-704	Treatment and Painting of Materiel.
MIL-P-15035	- Plastic Sheet: Laminated, Thermosetting, Cotton-Fabric-Base, Phenolic-Resin.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3920

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MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-52950	- Crates, Wood, Open and Covered.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of US Military Property.
MIL-STD-889	Dissimilar Metals.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ME

TA13217E1910	- Sled, Scow-Type, Plastic, 200 Pound (90 Kilogram) Capacity.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for commercial Packaging.

(Application for copies should be addressed to The American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

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(Application for copies should be addressed to the society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using 'Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The scow-type sled (hereinafter called "sled") shall be in accordance with TA13217E1910 and as specified herein. The sled shall consist of the hull, with three runners, each 3/4 inches (19.1 millimeters) wide (one on each side and one in the center) extending the length of the sled and attached directly to the platform, a cover, and a U-bolt for towing. The sled shall be capable of supporting a 200-pound (90 kilogram.) load equally distributed over the sled when tested as specified in 4.5.2.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the sled. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Approval of the first article will not relieve the contractor of his obligation to furnish sleds conforming to this specification.

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.3.1 Material deterioration and control. The sled shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the sled may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such

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as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.1.3 Plastic for runners and plugs. The plastic for the runners and plugs shall be cotton fabric base phenolic-resin, laminated, conforming to MIL-P-15035, type FBM.

3.3.1.4 Hardware. Unless otherwise specified herein, all hardware for the sled shall conform to TA13217E191O.

3.4 Runners and plugs. The runners and plugs shall be fabricated from plastic as specified in 3.3.1.3 and may be molded or cut from sheet or strip in the size and shape shown on the applicable drawing. The bottom of the runners shall be smooth. Any saw marks or rough surface shall be removed prior to attaching to the hull. Holes, counterbored to receive screwheads shall be filled with plugs constructed from plastic as specified in 3.3.1.3 pressfitted or cemented in a manner to prevent loss in use. Maximum gap allowance between hull and runner between screws shall be 1/32 inch (0.8 millimeters).

3.4.1 Battens. The cover when assembled to the hull by battens shall not have a gap greater than 3/32 inch (2.4 millimeters) between hull and canvas throughout full width of the battens and at any point around the hull. Battens shall be attached to the hull so that the vertical distance downward from the top of adjacent gunwale to highest edge of batten surmounted by one thickness of cover material, shall not be greater than 3/16 inch (4.8 millimeters) at any point around the sled. The screw hole through the batten shall be located 3/8 inch (9.5 millimeters) $\pm$ 1/16 inch (1.6 millimeters) from each edge.

3.5 Identification marking. The sled and cover shall be identified in accordance with MIL-STD-130. The marking for the sled shall be applied to the hull as specified on the drawings, The marking for the cover shall be applied by label in accordance with the drawings.

3.6 Workmanship.

3.6.1 Castings and forgings. All parts, components, and assemblies of the sled which include castings and forgings shall be clean of harmful extraneous material such as sands dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.6.2 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or

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self-locking nuts shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than two full threads.

3.6.3 External surfaces. External surf-aces shall be free of burrs, sharp edges and corners except where sharp edges or corners are required or Where they are not detrimental to safety.

3.6.4 Treatment and painting. The portions of the sled normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type F for ferrous metals and type G for non-ferrous metals and fiberglass, top coat color camouflage green 383 conforming to MIL-C-46168. Parts and components previ-ously primed by a subcontractor with alkyd base primer (color immaterial) may be used if cured for a minimum of 14 days and then cleaned and painted as specified herein either prior to or at final sled assembly painting. If another color is specified (see 602), the above system shall be overcoated with MIL-C-46168 of the desired color.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the preformance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings.

4.1.2 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on applicable drawings. The drawings specify the characteristics requiring inspection, the sampling plan, and the basis for acceptance and rejection (see 6.5 and 6.6).

4.1.3 Disassembly inspection. Failure of any examination or test by the first article model shall be cause for disassembly, in the presence of a Government representative, of the first article model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article model. Reassembly with replacement parts and retesting shall be the responsiblilty of the contractor.

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4.2 Classification of inspections. The inspection requirements specified herein are classifed as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article sled shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article sled shall be tested as specified in 4.5.2. Failure of the test shall be cause for performing the disassembly inspection specified in 4.1.3.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, level II.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Test. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. Failure to meet the test requirements of 4.5.2 shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The sled shall be examined for the characteristics in table I as specified in 4.4.2.

Table I. Examination schedule.

Number	Characteristics	Requirement paragraph
Major		
101.	The scow-type sled not in accordance with TA13217E1910.	3.1
102.	Material not as specified.	3.3
103.	Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage or operating environment.	3.3.1
104.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.1.1

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Table I. Examination schedule. (Continued).

Number	Characteristics	Requirement paragraph
105.	Contractor does not have documentation available for identification of material, material finishes or treatment.	3.3.1.2
106.	Plastic for runners and plugs not as specified.	3.3.1.3
107.	Hardware not as specified.	3.3.1.4
108.	Runners and plugs not as specified.	3.4
109.	Battens not as specified.	3.4.1
110.	Treatment and painting not as specified.	3.6.4
Minor		
201.	Identification marking not as specified.	3.5
202.	Workmanship shoddy, not free of burrs, slivers, sharp edges; not clean or free of other harmful extraneous material such as rust or corrosion.	3.6, 3.6.1, 3.6.2, 3.6.3

4.5.2 Test. Load the sled with 200 pounds (90 kilograms) of weight equally distributed over length of the sled and pull the weighted sled over snow or loose sand a distance of 200 feet (60 meters) using two single-trace type harnesses Repeat the test using one double-trace harness. Evidence of permanent deformation, damage or breaking of any component or failure of the sled to complete the test or nonconformance to 3.1 shall constitute failure of the test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for characteristics listed in table II. AQL shall be 2.5 percent defective.

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Table II. Examination schedule.

Number	Characteristic	Level A	Commercial
111.	Sleds not preserved as specified.	5.1.1	5.1.2
112.	Sleds not nested in quantity specified.	5.2.1.1	5.2.2
113.	Shipping container not as specified.	5.2.1.1	5.2.2
114.	Beveled wood skids not applied to shipping container as specified.	5.2.1.1	
115.	Sleds not supported as specified.	5.2.1.1	
116.	Strapping not as specified.	5.2.1.1	
117.	Marking not as specified.	5.3.1	5.3.2

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit pack. The inside lashing line of each sled shall be neatly coiled, tied in the center with cord or twine, and the coil secured in the bottom of the sled. The sled cover and the outside lashing rope shall be installed on the sled. The cover shall be smoothly folded to the bottom of the sled and the lashing rope installed sufficiently loose to allow nesting of the sleds. The excess rope shall be threaded through the sled D-rings and the lashing hook at the end of the rope fastened securely or tied with cord or twine to one of the D-rings.

5.1.2 Commercial. Commercial preservation shall be in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Unitization. Four sleds, each preserved as specified in 5.1, shall be nested together and packed in a crate conforming to MIL-C-52950, style A, type II except that the top and bottom crossmembers shall be nominal 1- by 6-inch (2.5- by 15-centimeters) lumber. Each crate shall be provided with not less than four nominal 3- by 4-inch (7.5- by 10-centimeters) wood skids attached with nails driven from inside the crate and positioned flatwise across the bottom, parallel to the ends of the crate, two flush with the outside edge

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and the other two under the intermediate crossmembers. Ends of the skids shall be beveled full depth at a 45-degree angle. The sleds shall be placed in the crate, bottoms up, with the lower-most sled resting on supporting members. Care shall be exercised that the canvas cover on the lowermost sled is not pinched between the gunwale of the sled and the supporting member. Blocking, bracing, and closure shall be in accordance with the appendix of the container specification Strapping shall be positioned over the crossrmembers adjacent to the skids and shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.2.2 Commercial. Four sleds, each preserved as specified in 5.1, shall be nested together and packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military. In addition to any special marking required by the contract or purchase order, level A marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with ASTM D 3951. Additionally, each shipping container shall be marked with the weight and cube.

6. NOTES

6.1 Intended use. The sled is intended for use as personnel and supply transportation on snow, to be drawn manually by means of either two single-trace type harness or one double-trace harness.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When first article is not required (see 3.2).
c. Degree of preservation and packing required (see 5.1 and 5.2).
do When finish A strapping is required (see 5.2.1.1).
e. Color required when other than as specified (see 3.6.4).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more sleds. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the document's first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

6.5 Quality assurance provision (QAP). The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a

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minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrective action taken when applicable.

6.6 Definitions.

6.6.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-11153F	Sled, Scow-Type, Plastic

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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